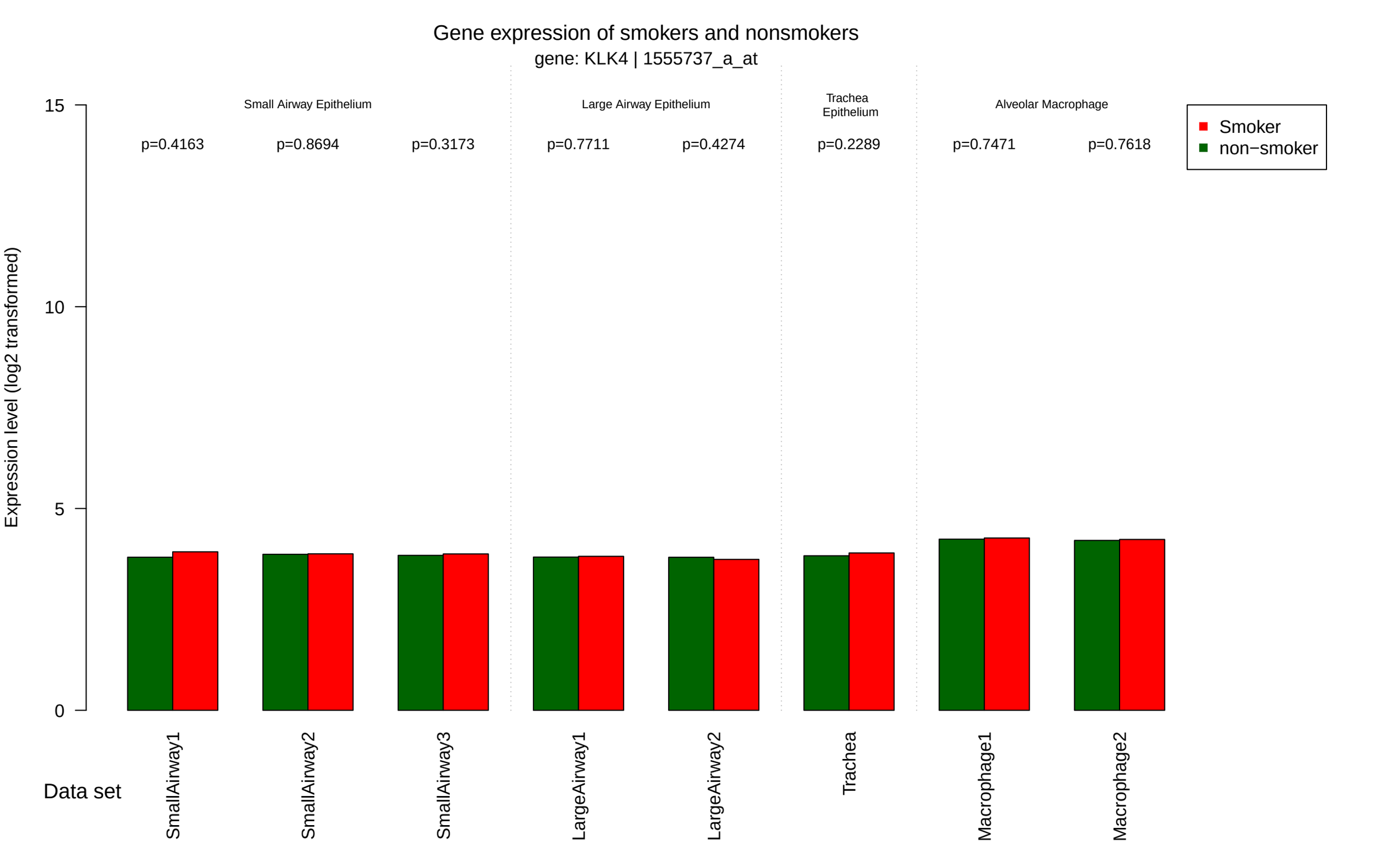
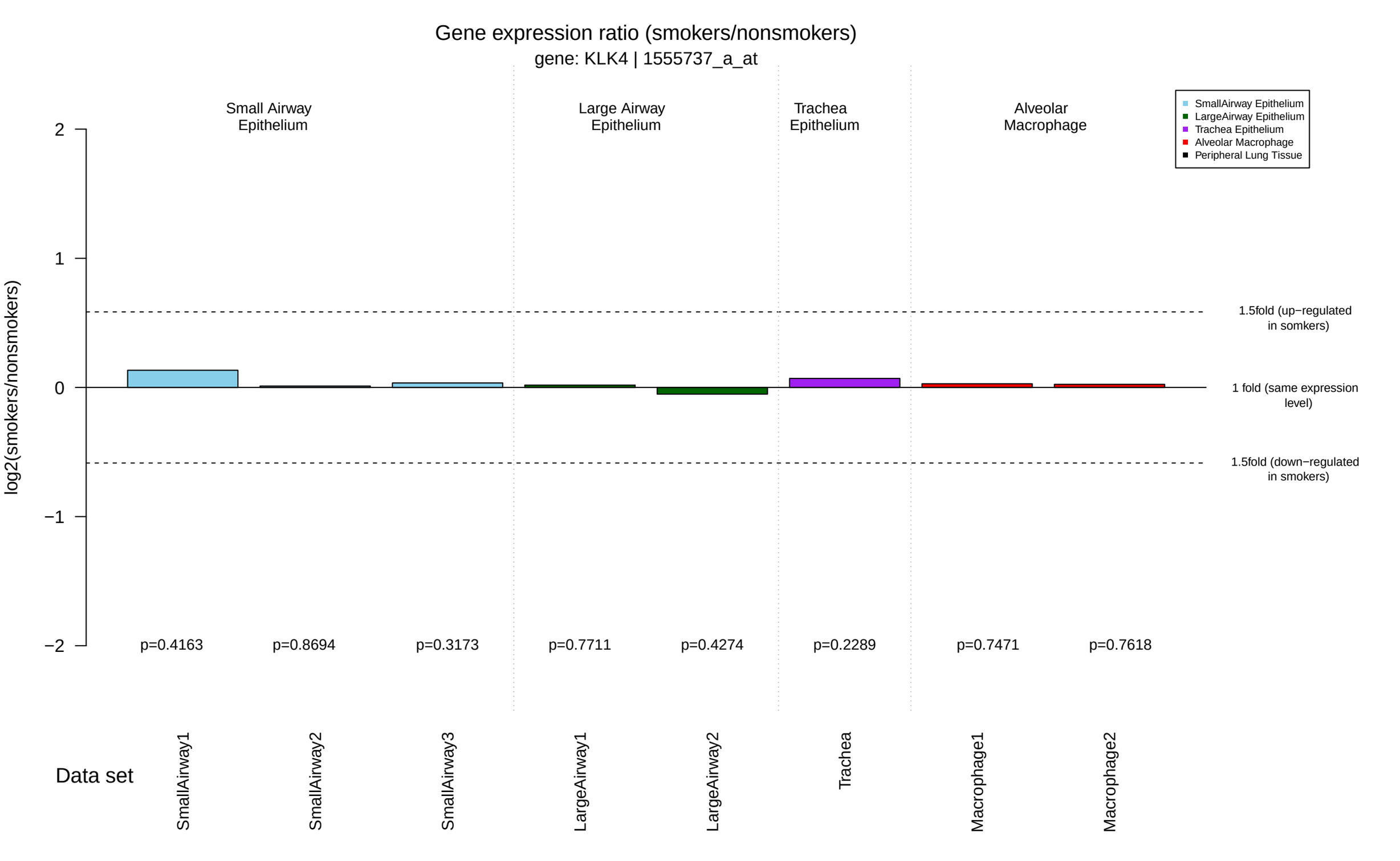
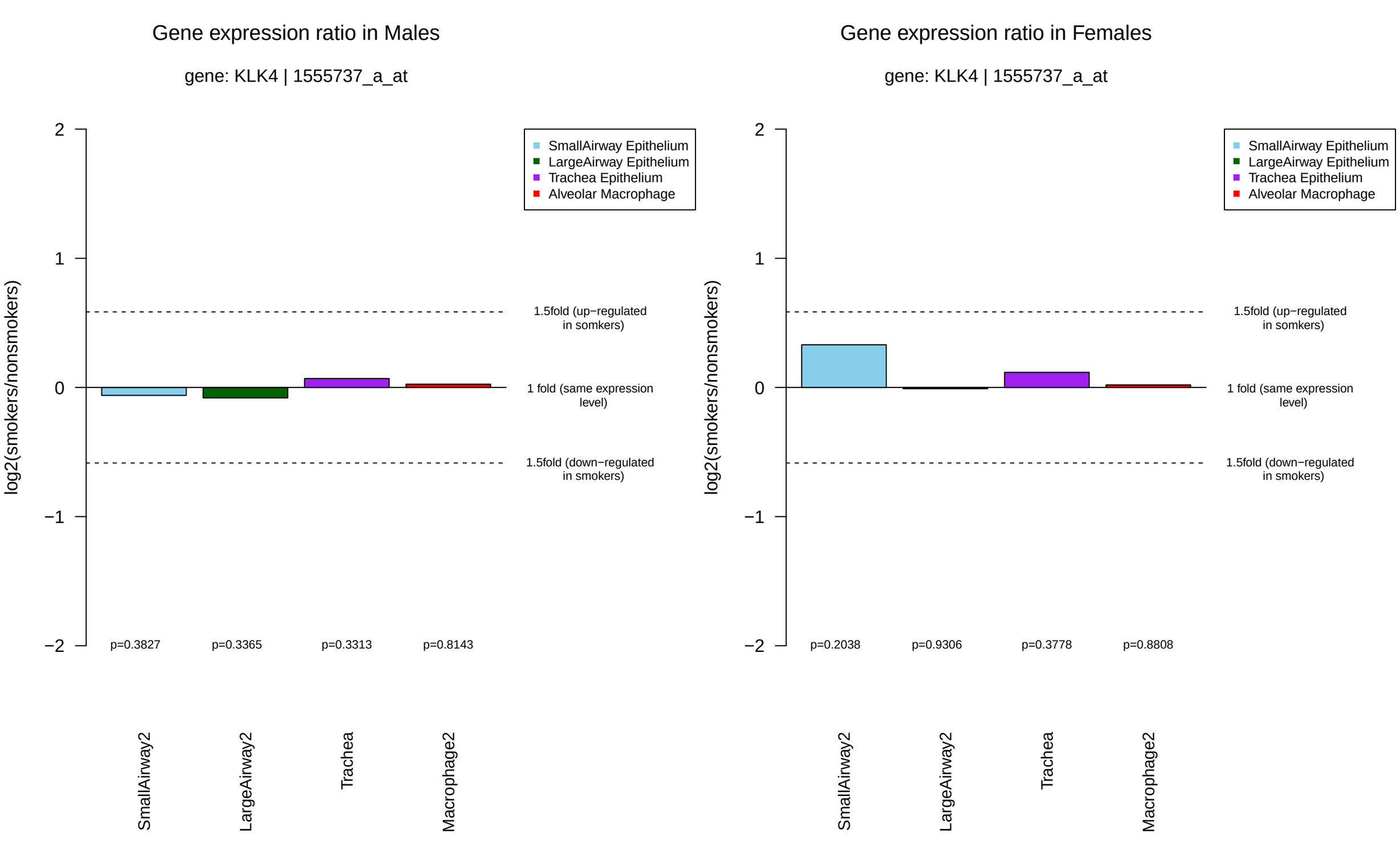
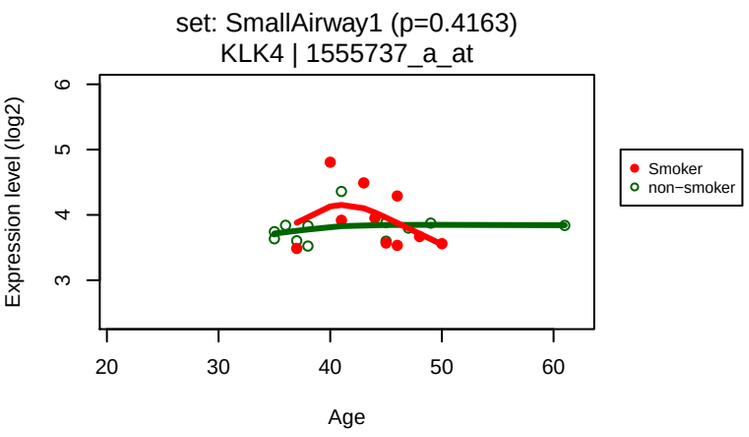
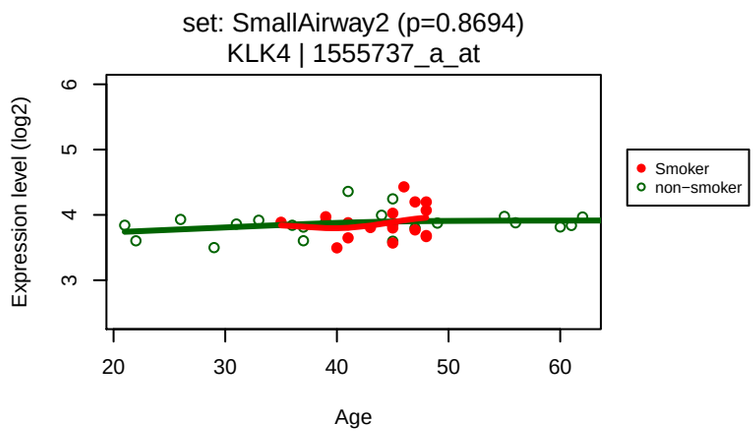
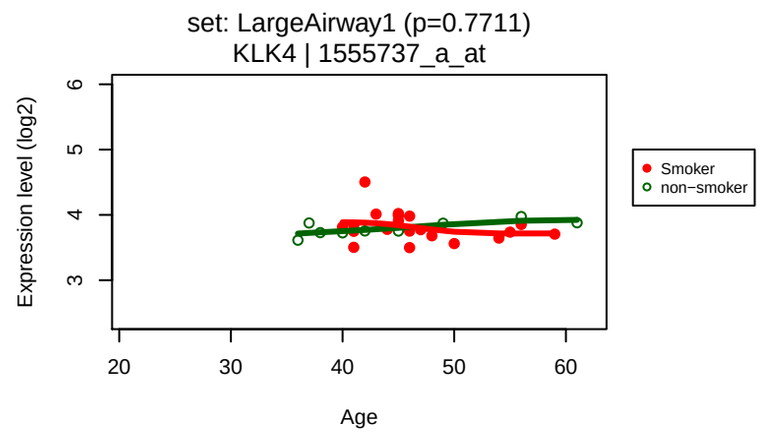

Gene expression of smokers and nonsmokers
gene: KLK4 | 1555737_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4163
p=0.8694
p=0.3173
p=0.7711
p=0.4274
p=0.2289
p=0.7471
p=0.7618
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

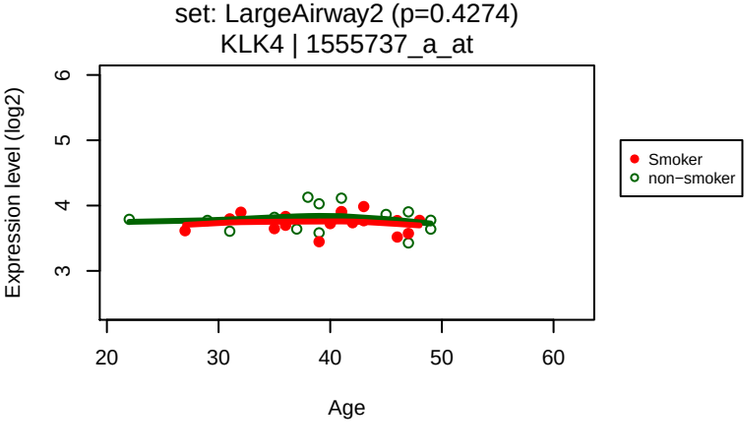

Gene expression ratio (smokers/nonsmokers)
gene: KLK4 | 1555737_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4163
p=0.8694
p=0.3173
p=0.7711
p=0.4274
p=0.2289
p=0.7471
p=0.7618
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

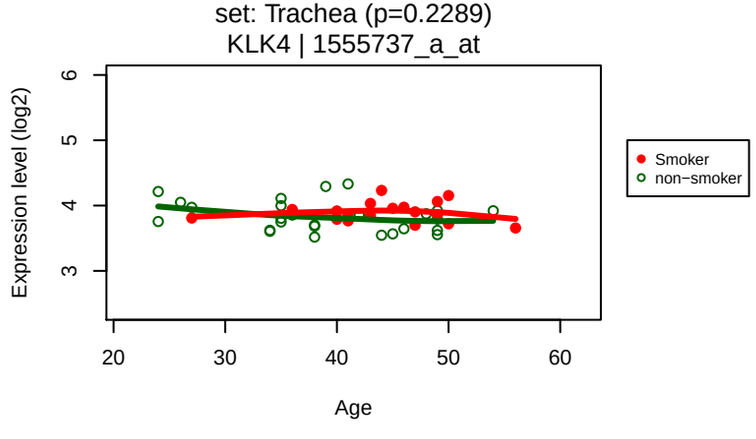

Gene expression ratio in Males
gene: KLK4 | 1555737_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3827
p=0.3365
p=0.3313
p=0.8143
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLK4 | 1555737_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2038
p=0.9306
p=0.3778
p=0.8808
SmallAirway2
LargeAirway2
Trachea
Macrophage2

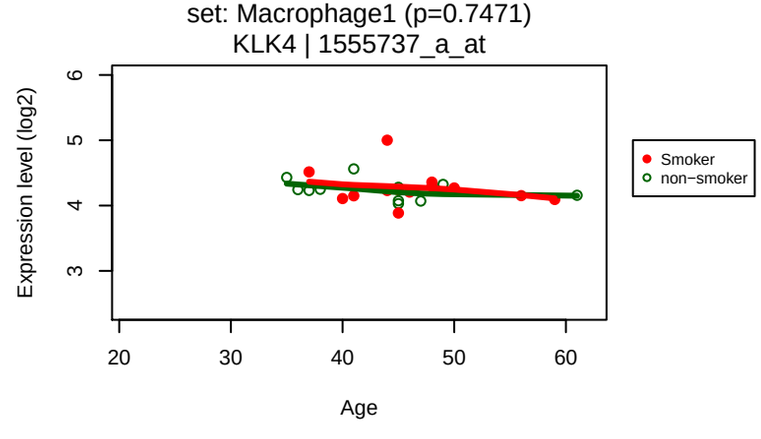

set: SmallAirway1 (p=0.4163)
KLK4 | 1555737_a_at
Expression level (log2)
Age
Smoker
non−smoker

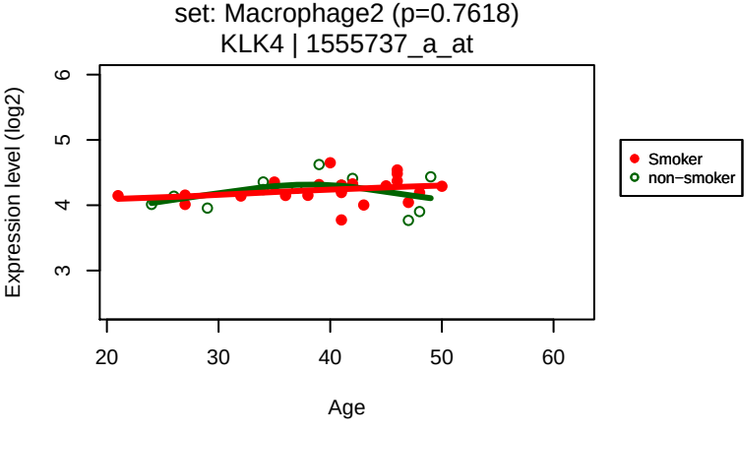

set: SmallAirway2 (p=0.8694)
KLK4 | 1555737_a_at
Expression level (log2)
Age
Smoker
non−smoker

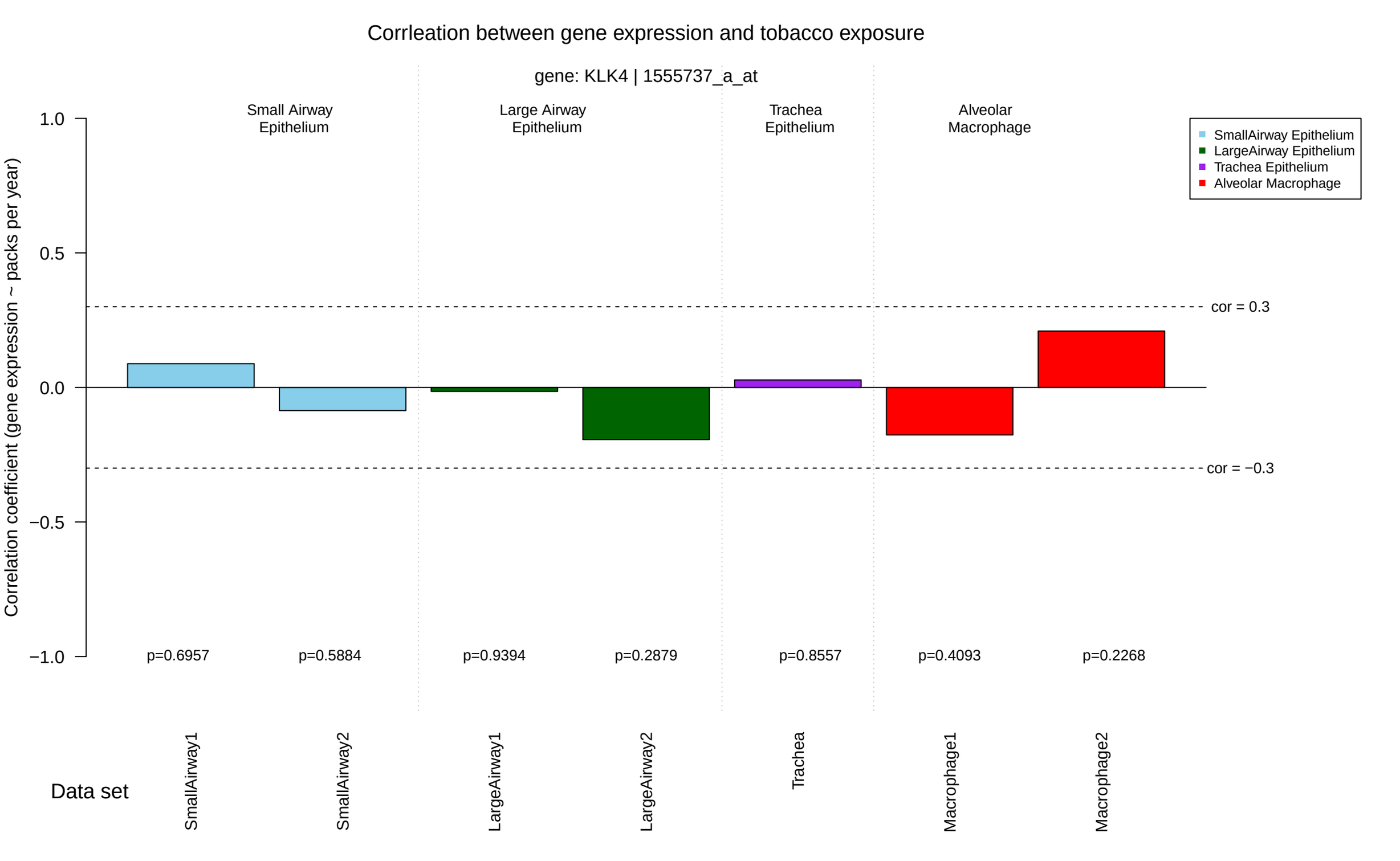

set: LargeAirway1 (p=0.7711)
KLK4 | 1555737_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4274)
KLK4 | 1555737_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2289)
KLK4 | 1555737_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7471)
KLK4 | 1555737_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7618)
KLK4 | 1555737_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLK4 | 1555737_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6957
p=0.5884
p=0.9394
p=0.2879
p=0.8557
p=0.4093
p=0.2268
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2